

POC Business Requirements Document (BRD)

Multi-Channel Delivery Configuration (MDC) — Campaign Field Setup

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1. Executive Summary

This **Proof of Concept (POC)** document defines the business requirements for configuring multi-channel delivery campaigns within the MDC (Multi-Channel Delivery Configuration) system. It focuses on the **data model** — the fields required to define a campaign, their relationships, dependencies, validation rules, and how they map to underlying data structures.

The POC supports four delivery channels: **SMS**, **EMAIL**, **PUSH**, and **LETTER**. A campaign configuration consists of 48 fields organized across 5 modules. Many fields are conditionally visible or required based on the selected delivery channels and the values of other fields. An AI Assistant guides business users through this complex field configuration process by intelligently sequencing prompts based on dependencies, pre-filling known values from reference use cases, and dynamically adapting questions to minimize conversation rounds while ensuring all required fields are captured.

POC Purpose: This document serves as a business alignment artifact to validate the field structure, relationships, and business rules before proceeding to full system development.

AI Assistant Focus: The POC emphasizes an AI Assistant that actively guides business users through the 48-field configuration process. Rather than presenting a static form, the assistant intelligently sequences field prompts based on dependencies, pre-fills known values from reference use cases, and dynamically adapts questions based on real-time user responses to minimize conversation rounds while ensuring all required fields are captured.

2. POC Scope & Objectives

POC Objectives

Objective	Description
AI Assistant Guidance	Demonstrate intelligent field prompting that reduces user effort through contextual, dependency-aware question sequencing
AI Assistant Feasibility	Demonstrate AI-assisted field configuration with minimal conversation rounds
Validate Data Model	Confirm the 48-field structure accurately represents business requirements
Validate Field Dependencies	Verify channel-driven and value-based field visibility logic
Validate Business Rules	Confirm validation rules and business constraints are correct

In Scope (POC)

- Field definitions, dependencies, and validation rules for all 5 modules
- AI-assisted configuration for module-by-module field completion
- Reference use case matching for field pre-population
- Channel-driven field visibility logic
- POC level UI/UX for demonstrating the campaign configuration workflow

Out of Scope (POC)

- Full production UI/UX implementation
- Integration with production MDC backend systems
- User authentication and role-based access control
- Audit logging and compliance reporting
- Multi-language support

3. Application Overview

Business Purpose

The MDC Campaign Configuration application enables business teams to define and register notification campaigns that deliver messages to customers across multiple channels. Each campaign configuration captures:

- **Who** owns and manages the campaign (people and departments)
- **What** type of message is being sent (classification, risk level)
- **Where** it applies (entity, market, line of business)
- **How** messages are delivered (channels, routing, sender identity)
- **What happens when delivery fails** (bounce back, fallback channels)

AI-Assisted Configuration (POC Capability)

The POC includes an intelligent assistant that guides users through the field configuration process. The assistant operates with the following business requirements:

- **Module-by-module guidance:** The assistant walks users through each module in sequence (Basic Info → Extension Info → Delivery Channel → Opt-In Flag → Bounce Back), ensuring all required fields are completed before advancing.
- **Minimal conversation rounds:** The assistant is designed to collect the maximum amount of information per interaction, reducing the number of back-and-forth exchanges needed to complete a full campaign configuration.
- **Contextual field prompting:** The assistant understands field dependencies and only prompts for fields that are relevant given the current configuration (e.g., it will not ask for PUSH-specific fields if no PUSH channel is selected).
- **Reference use cases:** The assistant can reference historical campaign configurations to pre-fill fields where patterns match, accelerating the setup process.

POC Note: The AI assistant demonstrates the feasibility of intelligent field guidance. Production implementation may include enhanced natural language understanding, integration with enterprise knowledge bases, and advanced pattern recognition.

4. Field Inventory by Module

Legend

Code	Meaning
RQ	Required — must be filled to proceed
RU	Required by UI — shown in the form, should be filled
CD	Conditional — required only when a dependency is met
OP	Optional — not required, user may fill if applicable
DP	Has dependencies on other fields or channels

4.1 Basic Info

Module Purpose: Captures the fundamental identity of the campaign — who it belongs to, what market it serves, and which system triggers it.

#	Field Name	Display Name	Business Description	Required	Options / Enum	Provided By	Dependencies
1	group_member	Entity	Entity / group member	RQ	HASE, HSBC	Business	—

#	Field Name	Display Name	Business Description	Required	Options / Enum	Provided By	Dependencies
2	country_code	Market	Business area the use case covers	RQ	INHK (HK/MO), HASE (HK only)	Business	—
3	line_of_business	Line Of Business	Line of business	RQ	WPB, RB, CMB	Business	DP Editable in create mode only
4	use_case_name	Use Case Name	Business scenario for message triggering; used for MDC MI report lookup	RQ	Free text	Business	—
5	project_name	Project Name	Project name	RQ	Free text	Business / IT PM	—
6	source_system	Source System	Source system that passes notification request to MDC	RQ	Free text	Business / IT PM / Source System IT	—
7	downstream_name	Downstream Name	Downstream system that MDC passes notification request to	OP	Free text	Business / IT PM / Downstream System IT	—
8	service_line	Service Line	Message servicing nature	RQ	Servicing, Marketing	Business	—

Total fields: 8 (7 Required, 1 Optional)

4.2 Extension Info

Module Purpose: Captures ownership hierarchy, message classification (risk, schedule, dual-channel), and business context (journeys, trigger conditions, regulatory requirements).

#	Field Name	Display Name	Business Description	Required	Options / Enum	Provided By	Dependencies
9	depart_head	Depart. Head	Department head, MD or GCB 2	RU	Free text	Business	—
10	team_head	Team Head	Team head, GCB 3	RU	Free text	Business	—
11	message_owner	Message Owner	Message owner, GCB 5 or above	RQ	Free text	Business	—
12	business_line_1st_level	Business Line 1st Level	Department of the department head	RU	Free text	Business	—
13	business_line_2nd_level	Business Line 2nd Level	Department of the team head	RU	Free text	Business	—
14	delivery_schedule	Is 7×24	Whether delivery schedule supports 7×24	RQ	Yes (7×24), No	Business	—
15	delivery_schedule_other	Other Schedule	Expected delivery frequency if not 7×24	RQ	Free text	Business	DP Depends on delivery_schedule = No
16	high_risk_flag	Is High Risk	High risk critical message flag (regulatory, time-sensitive, high-risk transaction)	RQ	Yes, No	Business	—
17	is_dual_channel	Is Dual Channel	Whether the message requires dual channel delivery	OP	Yes, No	Business	—
18	support_dual_vendor	Support Dual Vendor	Whether SMS requires dual vendor routing; mandatory for high-risk messages	OP	Yes, No	Business	DP Business rule: if high_risk_flag = Yes , then must be Yes
19	business_team	Business Team	Business team	OP	Free text	Business	—

#	Field Name	Display Name	Business Description	Required	Options / Enum	Provided By	Dependencies
20	business_contact	Business Contact	Business contact	OP	Free text	Business	—
21	message_trigger_conditions	Message Trigger Conditions	Message trigger conditions	OP	Free text (long)	Business	—
22	message_journey	Message Journey	Message journey	OP	Free text	Business	—
23	customer_journey	Customer Journey	Customer journey	OP	Free text	Business	—
24	business_journey	Business Journey	Business journey	OP	Free text	Business	—
25	remarks	Remarks	Key information	OP	Free text (long)	Business / IT (if any)	—
26	regulatory_requirement	Regulatory Requirement	Regulatory requirement details (e.g., MECP B9 + B11, TM-E-1 FAQ 4.1 Q11)	OP	Free text	Business	—
27	cost_owner	Cost Owner	Cost owner	RU	Free text	Business	—

Total fields: 19 (4 Required, 5 Required by UI, 10 Optional)

4.3 Delivery Channel

Module Purpose: Defines which channels are used for message delivery and configures channel-specific settings (sender identity, routing, encryption, traffic distribution).

#	Field Name	Display Name	Business Description	Required	Options / Enum	Provided By	Dependencies
28	channel	Delivery Channel	Selected delivery channel(s)	RQ	SMS , EMAIL , PUSH , LETTER	Business	—
29	priority	Priority	Channel priority / mandatory routing	CD	High , Medium , Low	Business	DP Required when any channel is selected
30	app_name	App Name	App name for Push Notification	CD	Free text	Business	DP Depends on PUSH channel
31	send_to_china_flag	Send to China	Whether to send to China mobile number	CD	Yes , No	Business	DP Depends on SMS channel
32	traffic_percentage	Traffic Percentage	Traffic split percentage across vendor paths	CD	Numeric (0–100)	Business / MDC IT	DP Depends on SMS , EMAIL , PUSH
33	sender	Sender	Sender ID/address/name depending on channel	CD	Free text	Business	DP Depends on SMS , EMAIL
34	sender_name	Sender Name	Email sender name	CD	Free text	Business	DP Depends on EMAIL
35	cost_center_id	Cost Center Id	SMS cost center ID	CD	Free text (numeric)	Business	DP Depends on SMS
36	encrypt_type	Encrypt Type	Email encryption type	CD	TLS (public/internal), Encrypt (restricted/highly restricted)	Business	DP Depends on EMAIL PFP path

Total fields: 9 (1 Required, 8 Conditional)

Typical Traffic Percentage Patterns:

Scenario	HTCL	CSL	Purpose
High-risk real-time message	100%	0%	Resilience
One-time-password (TC)	70%	30%	Resilience
Real-time / batch message	100%	—	Standard

Sender ID Examples by Channel:

Channel	Sender Format	Example
SMS	Sender ID number	e.g., HASE
EMAIL	Sender domain (bold part self-identifiable)	hangseng@mail.test. hangseng .com

4.4 Opt-In Flag

Module Purpose: Manages customer opt-in preferences for push notification channels. These flags determine whether a customer has consented to receive specific types of push notifications.

#	Field Name	Display Name	Business Description	Required	Options / Enum	Provided By	Dependencies
37	push_optin_flag	Master	Opt-in master flag	OP	Boolean	Business	DP Depends on PUSH channel
38	marketing_optin_flag	Marketing	Push marketing opt-in flag	OP	Boolean	Business	DP Depends on PUSH + app_name
39	high_risk_push_optin_flag	High Risk	Push high risk opt-in flag	OP	Boolean	Business	DP Depends on PUSH DAASC

Total fields: 3 (all Optional / Generated)

Opt-In Flag Hierarchy:

```

Master Opt-in (push_optin_flag)
├─ Marketing Opt-in (marketing_optin_flag)
└─ High Risk Opt-in (high_risk_push_optin_flag)

```

Note: Opt-In Flag fields are only relevant when the **PUSH** channel is selected. They are auto-generated based on channel and sub-path selections.

4.5 Bounce Back

Module Purpose: Configures fallback behavior when message delivery fails or does not receive confirmation within a specified period. Bounce back enables automatic routing to alternative channels.

#	Field Name	Display Name	Business Description	Required	Options / Enum	Provided By	Dependencies
40	bounce_back	Callback	Bounce back callback flag; mobile calls MDC API on PN receipt; triggers next channel if no callback within pre-set period	RU	Yes, No	Business	DP Depends on PUSH channel
41	letter_bounce_back_success_flag	Letter Bounce Back	For INHK eStatement/eAdvice: triggers e-Notification when eStatement/eAdvice is ready on website	OP	Yes, No	Business	DP Depends on LETTER channel
42	push_bounce_back_period	Push Bounce Back Period	Minutes to wait for PUSH callback before triggering next channel	OP	Numeric (minutes)	Business	DP Depends on bounce_back = Yes

#	Field Name	Display Name	Business Description	Required	Options / Enum	Provided By	Dependencies
43	sms_bounce_back_period	SMS Bounce Back Period	Minutes to wait for SMS SENT status before triggering next channel	OP	Numeric (minutes)	Business	DP Depends on unknown_bounce_back_status
44	email_bounce_back_period	Email Bounce Back Period	Minutes to wait for EMAIL SENT status before triggering next channel	OP	Numeric (minutes)	Business	DP Depends on unknown_bounce_back_status
45	letter_bounce_back_period	Letter Bounce Back Period	Minutes to wait for LETTER FAILED status before triggering next channel	OP	Numeric (minutes)	Business	DP Depends on unknown_bounce_back_status
46	bounce_back_next_channel	Bounce Back Next Channel	Whether to route to next channel after bounce back failure	OP	Yes, No	Business	—
47	unknown_bounce_back_status	Unknown Bounce Back Status	When enabled, MDC checks bounce back periods and triggers next channel if no successful SENT status received	OP	Yes, No	Business	—
48	auto_bounce_back_flag	Auto Update Invalid Flag	Auto-update mobile/email invalid flag to CUS when SMS/Email delivery fails with specific failure reasons	OP	Yes, No	Business	—

Total fields: 9 (1 Required by section, 8 Optional)

5. Field Dependencies & Relationships

This section documents all dependency relationships between fields, channels, and modules.

5.1 Channel-Driven Field Visibility

When a user selects one or more delivery channels, specific fields become visible or required across multiple modules.

Channel → Activated Fields Matrix

Field	SMS	EMAIL	PUSH	LETTER
Delivery Channel				
priority	RQ	RQ	RQ	RQ
send_to_china_flag	RQ			
traffic_percentage	RQ	RQ	RQ	
sender	RQ	RQ		
sender_name		RQ		
cost_center_id	RQ			
encrypt_type		RQ		
app_name			RQ	
Opt-In Flag				
push_optin_flag			RQ	
marketing_optin_flag			RQ	

Field	SMS	EMAIL	PUSH	LETTER
high_risk_push_optin_flag			RQ	
Bounce Back				
bounce_back			RQ	
sms_bounce_back_period	RQ			
email_bounce_back_period		RQ		
push_bounce_back_period			RQ	
letter_bounce_back_success_flag				RQ
letter_bounce_back_period				RQ

Fields Activated by Channel

Channel	Activated Fields
SMS	priority, send_to_china_flag, sender, cost_center_id, traffic_percentage, sms_bounce_back_period
EMAIL	priority, sender, sender_name, encrypt_type, traffic_percentage, email_bounce_back_period
PUSH	priority, app_name, traffic_percentage, push_optin_flag, marketing_optin_flag, high_risk_push_optin_flag, bounce_back, push_bounce_back_period
LETTER	priority, letter_bounce_back_success_flag, letter_bounce_back_period
All Channels	priority, traffic_percentage (SMS/EMAIL/PUSH only)

5.2 Cross-Field Value Dependencies

Some fields become visible, required, or change behavior based on the **value** of another field.

Value Dependency Matrix

Target Field	Depends On	Condition	Effect
delivery_schedule_other	delivery_schedule	= No	Field becomes visible/required
support_dual_vendor	high_risk_flag	= Yes	Must be Yes (business rule)
push_bounce_back_period	bounce_back	= Yes	Field becomes visible
sms_bounce_back_period	unknown_bounce_back_status	= Yes	Field becomes visible
email_bounce_back_period	unknown_bounce_back_status	= Yes	Field becomes visible
letter_bounce_back_period	unknown_bounce_back_status	= Yes	Field becomes visible
marketing_optin_flag	app_name	Has value	Field becomes visible
line_of_business	(create mode)	Always	Editable only in create mode

Value Dependency Rules

1. Delivery Schedule:

Condition	Result
delivery_schedule = Yes (7×24)	No additional field needed
delivery_schedule = No	delivery_schedule_other becomes required (specify delivery frequency)

2. High Risk → Dual Vendor:

Condition	Result
high_risk_flag = Yes	support_dual_vendor must be Yes (business rule: high-risk messages require dual vendor)
high_risk_flag = No	support_dual_vendor is optional

3. Bounce Back Period:

Channel	Condition	Result
PUSH	<code>bounce_back = Yes</code>	<code>push_bounce_back_period</code> becomes visible
SMS	<code>unknown_bounce_back_status = Yes</code>	<code>sms_bounce_back_period</code> becomes visible
EMAIL	<code>unknown_bounce_back_status = Yes</code>	<code>email_bounce_back_period</code> becomes visible
LETTER	<code>unknown_bounce_back_status = Yes</code>	<code>letter_bounce_back_period</code> becomes visible

4. Opt-In Flag Hierarchy (PUSH channel only):

Level	Field	Condition
1	<code>push_optin_flag</code>	Always visible when PUSH selected
2	<code>marketing_optin_flag</code>	Visible when PUSH selected AND <code>app_name</code> has value
3	<code>high_risk_push_optin_flag</code>	Visible when PUSH selected (DAASC path)

5.3 Cross-Module Dependencies

Fields in one module can affect fields in another module.

Module → Module Dependencies

Source Module	Source Field	Target Module	Target Field	Relationship
Basic Info	<code>service_line</code>	Extension Info	<code>regulatory_requirement</code>	Recommended if Servicing
Extension Info	<code>high_risk_flag</code>	Extension Info	<code>support_dual_vendor</code>	Must be Yes if high risk
Extension Info	<code>delivery_schedule</code>	Extension Info	<code>delivery_schedule_other</code>	Required if not 7×24
Delivery Channel	<code>channel</code>	Bounce Back	Various periods	Channel selection activates bounce back fields
Delivery Channel	<code>channel</code>	Opt-In Flag	Various opt-in flags	PUSH activates opt-in flags
Bounce Back	<code>bounce_back</code>	Opt-In Flag	<code>push_optin_flag</code>	Activates push opt-in

Channel → Cross-Module Field Activation

Channel	Delivery Channel Fields	Opt-In Flag Fields	Bounce Back Fields
SMS	<code>sender</code> , <code>send_to_china_flag</code> , <code>cost_center_id</code> , <code>traffic_percentage</code>	—	<code>sms_bounce_back_period</code>
EMAIL	<code>sender</code> , <code>sender_name</code> , <code>encrypt_type</code> , <code>traffic_percentage</code>	—	<code>email_bounce_back_period</code>
PUSH	<code>app_name</code> , <code>traffic_percentage</code>	<code>push_optin_flag</code> , <code>marketing_optin_flag</code> , <code>high_risk_push_optin_flag</code>	<code>bounce_back</code> , <code>push_bounce_back_period</code>
LETTER	—	—	<code>letter_bounce_back_success_flag</code> , <code>letter_bounce_back_period</code>

6. Data Architecture

Database Tables Overview

The 48 fields are stored across **3 database tables**:

Table	Purpose	Field Count	Relationship
<code>tbl_use_case</code>	Core campaign identity, classification, and opt-in flags	16	Primary record; one per campaign
<code>tbl_use_case_ext</code>	Extended details: ownership, journeys, business context	19	1:N with <code>tbl_use_case</code>
<code>tbl_use_case_channel_rule</code>	Channel-specific delivery configuration	9	1:N with <code>tbl_use_case</code> ; one rule per channel

Table: `tbl_use_case` (Core Campaign Identity)

Field Name	Display Name	Description
group_member	Entity	HASE or HSBC
country_code	Market	INHK (HK/MO) or HASE (HK only)
line_of_business	LOB	WPB, RB, or CMB
use_case_name	Use Case Name	Campaign name
project_name	Project Name	Project identifier
source_system	Source System	Originating system
service_line	Service Line	Servicing or Marketing
high_risk_flag	Is High Risk	Yes/No
delivery_schedule	Is 7x24	7x24 or custom schedule
is_dual_channel	Is Dual Channel	Dual channel flag
bounce_back	Callback	Bounce back callback flag
letter_bounce_back_success_flag	Letter BB	Letter bounce back flag
push_optin_flag	Master Opt-in	Opt-in master flag
marketing_optin_flag	Marketing Opt-in	Push marketing opt-in
high_risk_push_optin_flag	High Risk Opt-in	Push high risk opt-in

Table: tbl_use_case_ext (Extended Campaign Details)

Field Name	Display Name	Description
downstream_name	Downstream	Downstream system name
depart_head	Depart. Head	Department head name
team_head	Team Head	Team head name
message_owner	Message Owner	Message owner name
business_line_1st_level	BL 1st Level	Business line first level
business_line_2nd_level	BL 2nd Level	Business line second level
business_team	Business Team	Business team name
business_contact	Business Contact	Business contact person
message_trigger_conditions	Trigger Conditions	Message trigger conditions
message_journey	Message Journey	Message journey type
customer_journey	Cust. Journey	Customer journey type
business_journey	Bus. Journey	Business journey type
remarks	Remarks	Additional notes
regulatory_requirement	Reg. Requirement	Regulatory requirements
cost_owner	Cost Owner	Cost owner name
support_dual_vendor	Dual Vendor	Support dual vendor flag

Table: tbl_use_case_channel_rule (Channel-Specific Configuration)

Field Name	Display Name	Description
channel	Delivery Channel	SMS, EMAIL, PUSH, or LETTER
priority	Priority	High, Medium, or Low
traffic_percentage	Traffic %	Traffic split percentage
sender	Sender ID/Addr	Sender ID or address
sender_name	Sender Name	Email sender name
cost_center_id	Cost Center	SMS cost center ID
encrypt_type	Encrypt Type	TLS or Encrypt

Field Name	Display Name	Description
send_to_china_flag	Send to China	China mobile number flag
app_name	App Name	Push notification app name

Relationships

- Each `tbl_use_case` can have **one** `tbl_use_case_ext` (1:1)
- Each `tbl_use_case` can have **multiple** `tbl_use_case_channel_rule` records (1:N), one per selected channel

7. Validation Rules & Business Rules

7.1 Field Requirement Rules

Rule ID	Field(s)	Rule Type	Rule Description
VR-01	group_member, country_code, line_of_business, use_case_name, project_name, source_system, service_line	Required	Must be filled before completing Basic Info module
VR-02	message_owner, delivery_schedule, high_risk_flag	Required	Must be filled before completing Extension Info module
VR-03	channel	Required	At least one delivery channel must be selected
VR-04	depart_head, team_head, business_line_1st_level, business_line_2nd_level, cost_owner	Required by UI	Displayed in form; strongly recommended to fill
VR-05	priority	Conditional	Required when any delivery channel is selected

7.2 Conditional Visibility Rules

Rule ID	Target Field	Condition	Behavior
CV-01	delivery_schedule_other	delivery_schedule = No	Field becomes visible and required
CV-02	app_name	PUSH channel selected	Field becomes visible and required
CV-03	send_to_china_flag	SMS channel selected	Field becomes visible
CV-04	sender	SMS or EMAIL channel selected	Field becomes visible
CV-05	sender_name	EMAIL channel selected	Field becomes visible
CV-06	cost_center_id	SMS channel selected	Field becomes visible
CV-07	encrypt_type	EMAIL channel selected (PFP path)	Field becomes visible
CV-08	traffic_percentage	SMS, EMAIL, or PUSH channel selected	Field becomes visible
CV-09	push_optin_flag	PUSH channel selected	Field becomes visible
CV-10	marketing_optin_flag	PUSH channel selected AND app_name has value	Field becomes visible
CV-11	high_risk_push_optin_flag	PUSH channel selected (DAASC path)	Field becomes visible
CV-12	bounce_back	PUSH channel selected	Field becomes visible
CV-13	push_bounce_back_period	bounce_back = Yes	Field becomes visible
CV-14	sms_bounce_back_period	SMS channel selected AND unknown_bounce_back_status = Yes	Field becomes visible
CV-15	email_bounce_back_period	EMAIL channel selected AND unknown_bounce_back_status = Yes	Field becomes visible
CV-16	letter_bounce_back_period	LETTER channel selected AND unknown_bounce_back_status = Yes	Field becomes visible
CV-17	letter_bounce_back_success_flag	LETTER channel selected	Field becomes visible

7.3 Business Logic Rules

Rule ID	Rule Name	Description	Affected Fields
BR-01	High-Risk Dual Vendor	If <code>high_risk_flag</code> = Yes, then <code>support_dual_vendor</code> must be Yes	<code>high_risk_flag</code> , <code>support_dual_vendor</code>
BR-02	High-Risk Traffic Split	High-risk real-time messages should use HTCL-100%, CSL-0% for resilience	<code>traffic_percentage</code>
BR-03	OTP Traffic Split	One-time-password messages should use HTCL-70%, CSL-30% for resilience	<code>traffic_percentage</code>
BR-04	Real-time/Batch Traffic	Standard real-time or batch messages should use HTCL-100%	<code>traffic_percentage</code>
BR-05	Bounce Back → Next Channel	If <code>bounce_back</code> or <code>unknown_bounce_back_status</code> is enabled, <code>bounce_back_next_channel</code> should be configured	<code>bounce_back_next_channel</code>
BR-06	Auto Invalid Flag	<code>auto_bounce_back_flag</code> controls whether failed delivery automatically marks customer contact as invalid in CUS	<code>auto_bounce_back_flag</code>
BR-07	Regulatory + Service Line	If <code>service_line</code> = <code>Servicing</code> , <code>regulatory_requirement</code> is recommended to be filled	<code>regulatory_requirement</code>
BR-08	PUSH Opt-in Master	If PUSH channel is selected, <code>push_optin_flag</code> (master) must be addressed	<code>push_optin_flag</code>
BR-09	Line of Business Edit Lock	<code>line_of_business</code> is only editable during campaign creation, not in edit mode	<code>line_of_business</code>

7.4 Value Constraint Rules

Rule ID	Field	Constraint	Description
VC-01	<code>traffic_percentage</code>	0–100	Must be a valid percentage
VC-02	<code>push_bounce_back_period</code>	Positive integer	Minutes; must be > 0
VC-03	<code>sms_bounce_back_period</code>	Positive integer	Minutes; must be > 0
VC-04	<code>email_bounce_back_period</code>	Positive integer	Minutes; must be > 0
VC-05	<code>letter_bounce_back_period</code>	Positive integer	Minutes; must be > 0
VC-06	<code>cost_center_id</code>	Numeric string	e.g., <code>25267613</code>

8. Field Metadata

8.1 Provided By Summary

Provider Role	Fields Count	Typical Fields
Business	40	Most fields; primary owner of campaign configuration
Business / IT PM	2	<code>project_name</code> , <code>downstream_name</code>
Business / IT PM / Source System IT	1	<code>source_system</code>
Business / IT PM / Downstream System IT	1	<code>downstream_name</code>
Business / MDC IT	1	<code>traffic_percentage</code>
Business / IT (if any)	1	<code>remarks</code>

8.2 DB Table Mapping Summary

Table	Module(s)	Field Count
<code>tbl_use_case</code>	Basic Info (partial), Extension Info (partial), Opt-In Flag, Bounce Back	16
<code>tbl_use_case_ext</code>	Basic Info (partial), Extension Info (partial)	19
<code>tbl_use_case_channel_rule</code>	Delivery Channel	9
(no table)	Extension Info (<code>delivery_schedule_other</code>)	1

8.3 Complete Field Count by Module

Module	Required	Required by UI	Conditional	Optional	Total
Basic Info	7	0	0	1	8

Module	Required	Required by UI	Conditional	Optional	Total
Extension Info	4	5	0	10	19
Delivery Channel	1	0	8	0	9
Opt-In Flag	0	0	0	3	3
Bounce Back	0	1	0	8	9
Total	12	6	8	22	48

9. POC Limitations & Future Considerations

POC Limitations

Area	Limitation	Impact
Data Persistence	POC uses mock data; no real database integration	Field values are not persisted between sessions
AI Assistant	Demonstrates feasibility only; limited training data	May not handle all edge cases or complex queries
Validation	Basic client-side validation; no server-side enforcement	Business rules are for demonstration purposes
Integration	No integration with production MDC, CUS, or other systems	Field values are illustrative only
User Management	No authentication or role-based access	All users have full access in POC

Future Considerations

Area	Recommendation
Production Data Model	Validate ERD with database team; consider indexing strategy
AI Training	Expand training data with real historical campaigns
Validation Engine	Implement server-side validation for all business rules
Integration	Design APIs for MDC, CUS, and downstream system integration
Audit Trail	Add change tracking for compliance requirements
Multi-language	Support Traditional Chinese, Simplified Chinese, English

End of POC BRD — Full Version